CATERPILLAR®

DEALER BEST PRACTICE SERIES

Contamination Control Covers During Components Removal and Installation

Maintenance and Repair | Application | **Component Life Management** | Component Renewal (CRC) | MARC Management

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1.0 Introduction

Components that must be removed due to equipment repair in the field (OHT, TTT, MG etc.) are likely to be affected by contamination due to the environmental conditions at the mine site. Leaving the component exposed to the environment can result in larger problems apart from those solved thru the repair.

2.0 Best Practice Description

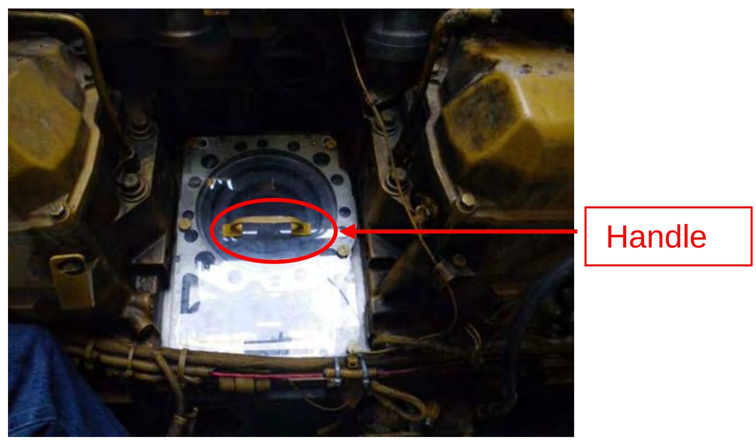
Considering continuous improvement and aiming at the contamination control of the work done at mine site, the use of a protector (acrylic) for the equipment repairs has been implemented. The cover protects the exposed area when the component is removed, improving the working conditions (reducing the risk of equipment damage).

Case 1: Cylinder Block Protection Cover

The picture shows the cylinder area exposed to the contamination during cylinder head removal and installation. The camshaft area is plugged with a wad of cloth that can cause the component to be exposed to additional contamination.

I. The acrylic protector enables visibility, and at the same time, it is linked together to a plate-spacer to cover the cylinders in the cylinder blocks of CAT® engines.

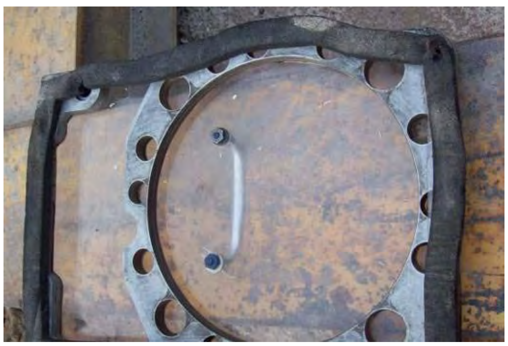
II. The protector has a handle that allows its manipulation and easy removal.



III. The protector is made in proportion to the plate-spacer in order to adjust perfectly in the housing. The protector is made by using a used plate-spacer.

The Plate-Spacer depends on the engine model.

IV. A seal keeps the piston / liner area from being contaminated.



Case 2: 793 Truck Housing Protection Cover

As part of the Planned Component Replacement (PCR) quality improvement program for a fleet of trucks it was necessary to execute a contamination control plan every time the final drive, transmissions exchange or repair regarding cracks or threads repair is done. As a result an acrylic cover with foamy seals was created.

3.0 Implementation Steps

- ✓ Existing need to fight contamination was determined.
- ✓ Protectors design: In Case 1 - taking the plate-spacer as a baseline for a determined engine. In Case 2 - the 793 truck final drive coupling dimensions were taken.
- ✓ Protector assembled along with the spacer and the handle.
- ✓ A seal installed to make sure that the protector would isolate exposed opening completely.
- ✓ Decision of implemented is made.

4.0 Benefits

- Because of the protector implementation the contamination inside the components is greatly reduced, if not eliminated, during the removal process. Additionally, the oil / coolant contamination is avoided resulting in the component being more reliable during its lifecycle.

- The economic benefits will be seen thru a longer equipment and/or component life, more working hours (availability), and less downtime.

5.0 Resources Required

- Acrylic Sheet
- Used Plate-Spacer
- Handle
- Gasket (Seal)

Case 1: If a new plate-spacer is used (Example: Plate-Spacer 110-6994) the cost in DBS is approximately $529000 COP ($285 USD). The acrylic, bolts, handle and the thread on the spacer cost will be approximately $60000 COP ($32 USD). The caps can be used on 8WM engines. The same concept can be applied on other engines; the proper plate-spacer must be used depending on each cylinder block.

Case 2: The manufacture cost is approximately $100000 COP ($54 USD). These cover are used in 793 trucks housings but the design can be adapted to use on any equipment to be repaired.

6.0 Supporting Attachments / References

None

7.0 Related Best Practices

1206-2.9-1048 Contamination Control Using Plugs & Caps
0911-4.03-1230 WIP Parts Storage Boxes and Lifting Tool

8.0 Acknowledgements

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